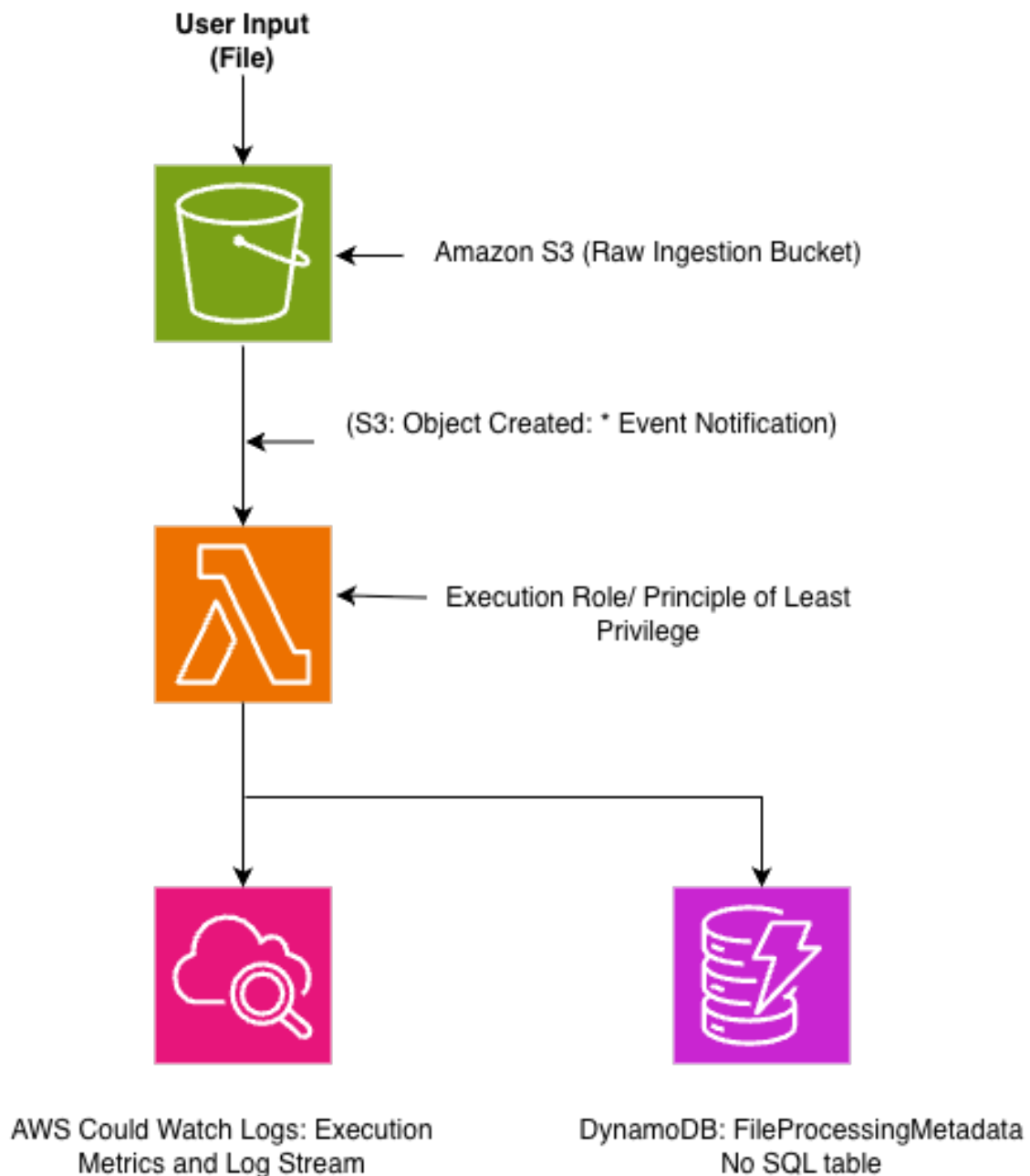


Building a Serverless Data Processing Pipeline on AWS

Cloud Services and Platforms - Graded Assignment

1. Architecture Diagram



2. Architectural Explanation

A. How Services Interact

- **Ingestion & Triggering:** The pipeline initiates when a user uploads a `.CSV` data file into the Amazon S3 raw data bucket. This upload triggers an asynchronous `S3:ObjectCreated:*` event notification.
- **Compute Execution:** Amazon S3 routes this event payload to the AWS Lambda function. The runtime instantiates, reads the object metadata (bucket name and file key), and downloads the CSV file payload directly into memory using the `boto3` client SDK.
- **Storage & Monitoring:** Lambda processes the file to calculate metrics (row count, column count, and size). It concurrently executes a `put_item` API call to push a structured metadata payload into the Amazon DynamoDB NoSQL table (`FileProcessingMetadata`) and writes the processed metrics directly to standard output, which Amazon CloudWatch Logs automatically captures via standard log streams.

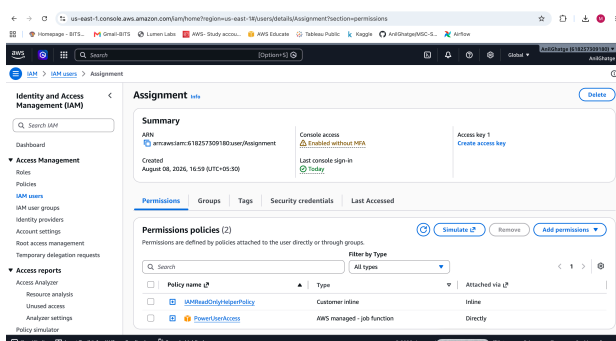
B. How Security is Enforced

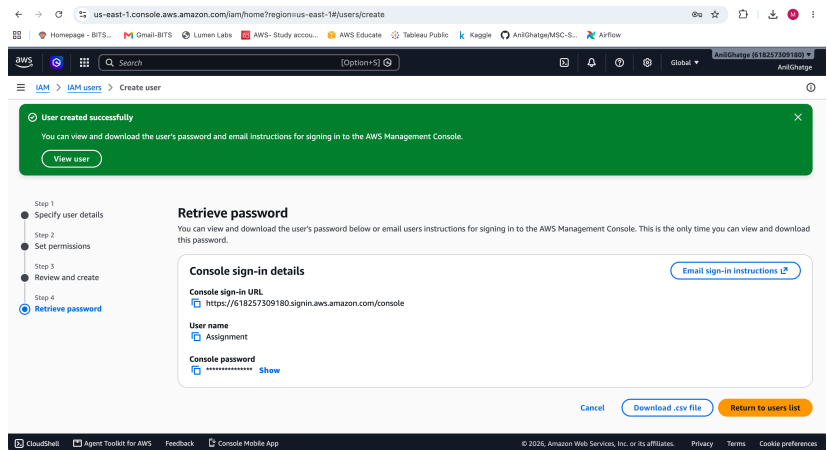
- **Identity & Access Management (IAM):** Security is strictly managed through the Principle of Least Privilege using decoupled, granular IAM roles.
- **User-Level Security:** The non-root `Assignment` IAM user handles provisioning and infrastructure deployment with restricted administrative capabilities.
- **Service-Level Security:** The AWS Lambda function executes under an attached execution role (IAM Service Role). This policy grants permission only to read objects from the specific S3 ingestion bucket (`s3:GetObject`), write logs to the specific CloudWatch Log Group (`logs:CreateLogStream`, `logs:PutLogEvents`), and insert unique records into the targeted DynamoDB table (`dynamodb:PutItem`). No external or unprivileged network traversal is permitted.

C. Why Event-Driven Design is Used

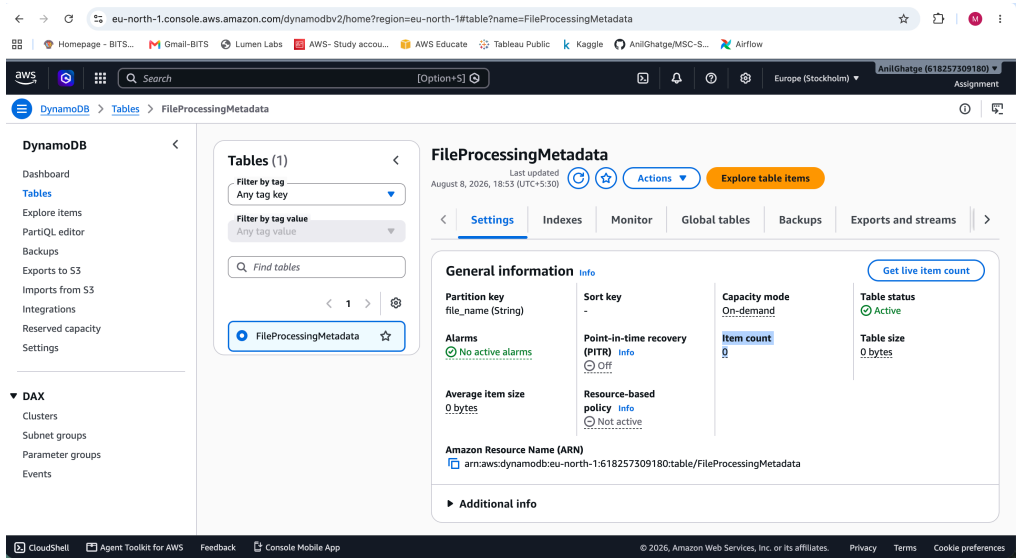
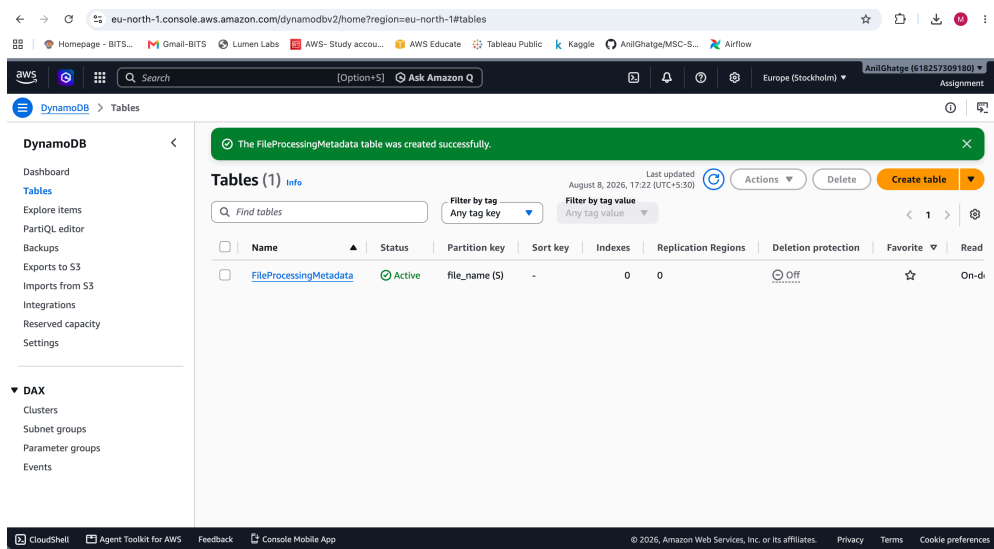
- **Cost Efficiency:** The entire architecture utilizes serverless resources that incur zero idling costs. Compute expenses are calculated purely on a per-millisecond execution duration when a file actually traverses the system.
- **Scalability & Resiliency:** The decoupled, asynchronous communication model between Amazon S3 and AWS Lambda ensures that spikes in data ingestion (e.g., hundreds of concurrent file uploads) are automatically scaled out smoothly by AWS managed infrastructure without crashing downstream applications or requiring manual server provisioning.

3. IAM User Creation





4. DynamoDB table creation



5. Lambda Function set up

The screenshot displays the AWS Lambda console for the function 'S3FileProcessor' in the 'eu-north-1' region. The 'Code' tab is active, showing a Python script for a Lambda handler. The script imports necessary libraries (json, boto3, datetime, csv, io, os) and initializes an S3 client and a DynamoDB resource. The handler function processes an event record by extracting bucket and key information, downloading the file from S3, and saving it to the local file system.

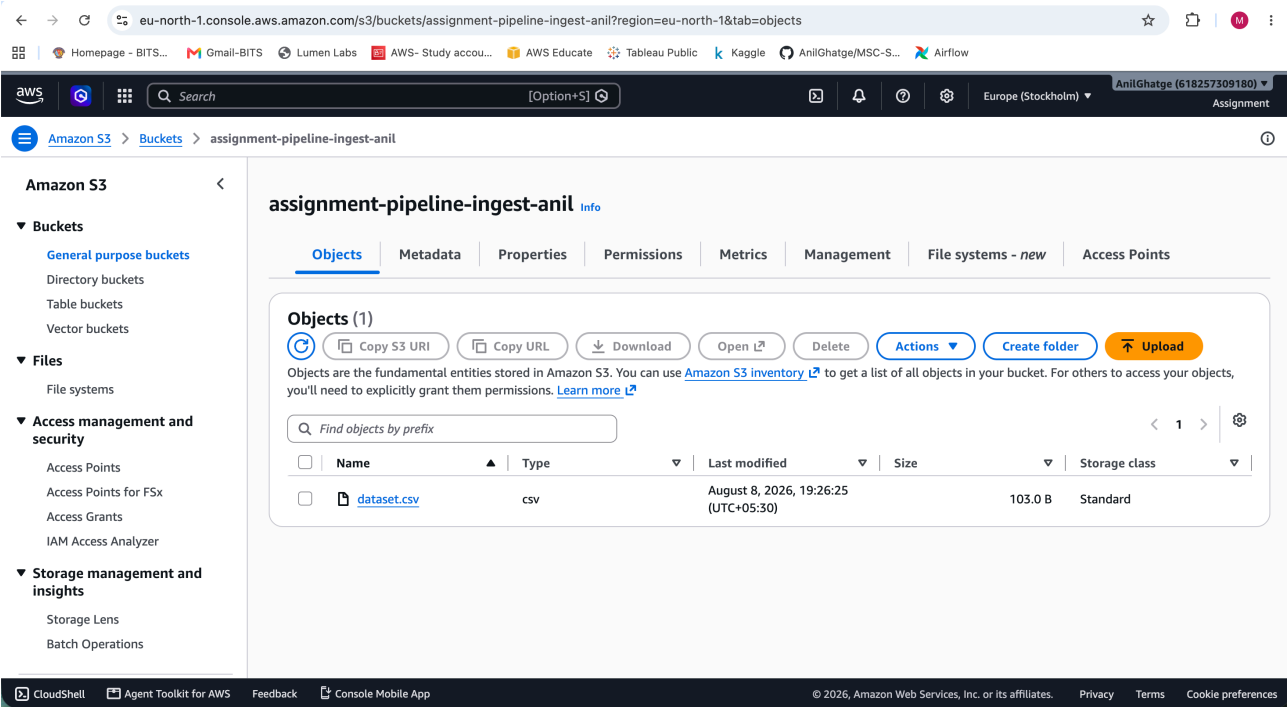
A green notification banner at the top indicates: "Successfully updated the function 'S3FileProcessor'." The right sidebar offers a 'Tutorials' section with a 'Create a simple web app' tutorial, which guides the user through building a web application using a Lambda function triggered by an S3 event.

The bottom section shows the 'Configuration' tab, where the function's overview is displayed. It includes the Function ARN, a description, and the last modified time. The 'Triggers' section shows a single trigger configured for S3, with the event source set to 'S3: assignment-pipeline-ingest-anil'.

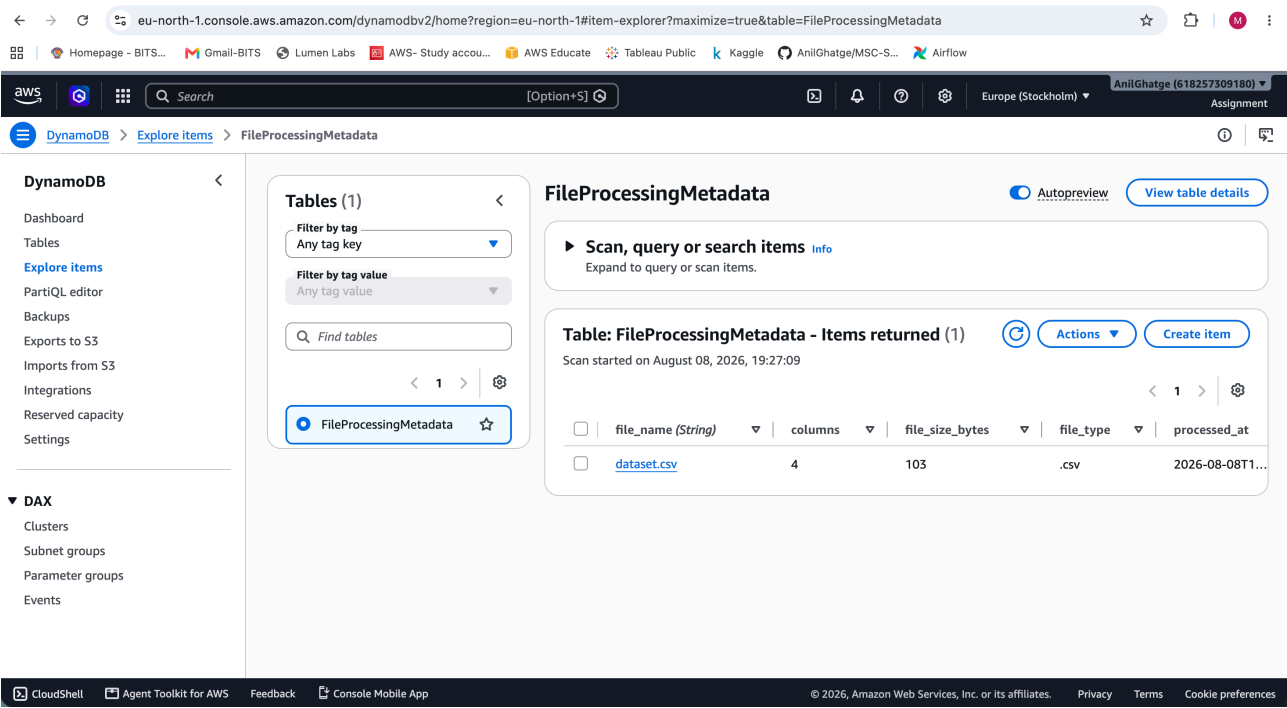
```
1 import json
2 import boto3
3 import datetime
4 import csv
5 import io
6 import os
7
8 s3_client = boto3.client('s3')
9 dynamodb = boto3.resource('dynamodb')
10 table = dynamodb.Table('FileProcessingMetadata')
11
12 def lambda_handler(event, context):
13     record = event['Records'][0]
14     bucket_name = record['s3']['bucket']['name']
15     file_key = record['s3']['object']['key']
16     file_size = record['s3']['object']['size']
17
```

6. Pipeline Functioning

File Loaded to S3 Bucket



File Loaded to DynamoDB



CloudWatch Log

eu-north-1.console.aws.amazon.com/cloudwatch/home?region=eu-north-1#logsV2:log-groups/log-group/\$252Faws\$252FIambda\$252FS3FileProcessor/log-events/202...

Homepage - BITS...Gmail-BITSLumen LabsAWS- Study accou...AWS EducateTableau PublicKaggleAnilGhatge/MSC-S...Airflow

awsSearch[Option+S]

Europe (Stockholm)AnilGhatge (618257309180)Assignment

CloudWatchLog managementaws/lambda/S3FileProcessor2026/08/08/[\$LATEST]707f01ed99134b2db3662988af455731

CloudWatch<

Favorites and recents▶

Ingestion

Dashboards

Alarms⚠️🕒🔍📉

▶ AI Operations

▶ GenAI Observability

▶ Application Signals (APM)

▶ Infrastructure Monitoring

▼ Logs

Log Management

Log AnalyticsNew

Log Anomalies

▶ Metrics

▶ Network Monitoring

▶ Setup

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear1m30m1h12hCustomUTC timezone

Display

Timestamp	Message
2026-08-08T13:49:39.028Z	END RequestId: beb48e5d-60f4-4178-9006-703b18c3eb50
2026-08-08T13:49:39.028Z	REPORT RequestId: beb48e5d-60f4-4178-9006-703b18c3eb50 Duration: 221.30 ms Billed Duration: 222 ms Memory Size: 1...
2026-08-08T13:51:39.175Z	START RequestId: beb48e5d-60f4-4178-9006-703b18c3eb50 Version: \$LATEST
2026-08-08T13:51:39.388Z	[ERROR] UnicodeDecodeError: 'utf-8' codec can't decode byte 0x97 in position 11: invalid start byte Traceback (mo...
2026-08-08T13:51:39.408Z	END RequestId: beb48e5d-60f4-4178-9006-703b18c3eb50
2026-08-08T13:51:39.408Z	REPORT RequestId: beb48e5d-60f4-4178-9006-703b18c3eb50 Duration: 232.13 ms Billed Duration: 233 ms Memory Size: 1...
2026-08-08T13:56:25.254Z	START RequestId: 4f58e337-cd6e-4be3-9643-dced90099f94 Version: \$LATEST
2026-08-08T13:56:25.498Z	Processed File Metrics: {"file_name": "dataset.csv", "file_type": ".csv", "file_size_bytes": 103, "rows": 3, "col...
2026-08-08T13:56:25.730Z	END RequestId: 4f58e337-cd6e-4be3-9643-dced90099f94
2026-08-08T13:56:25.730Z	REPORT RequestId: 4f58e337-cd6e-4be3-9643-dced90099f94 Duration: 475.87 ms Billed Duration: 476 ms Memory Size: 1...

No newer events at this moment. Auto retry paused. [Resume](#)

Back to top

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App© 2026, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences